

L3. Spins, Qubits, and Linear Operators for Measurement

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Quantum Mechanics Is Simply Different

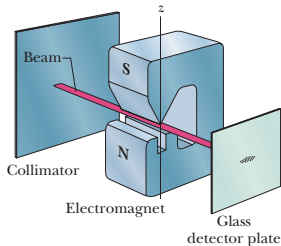
- Quantum mechanical objects are difficult to perceive with our sensory organs. For example, a microscopic particle, such as an electron, is not completely specified by its location. We need to understand particles and their motion as mathematical abstractions.
- The idea of a state in quantum mechanics is conceptually different from its classical counterpart. Unlike in the classical world, in which one can perform an experiment to determine the state of a system, state and measurement in the quantum world are two different things.

Outline

- The Stern-Gerlach Experiment
- Spin, Qubit, and the Bloch Sphere
- Linear Operators and Matrices
- The Principles of Quantum Mechanics (1-4) and Their Applications on Spin

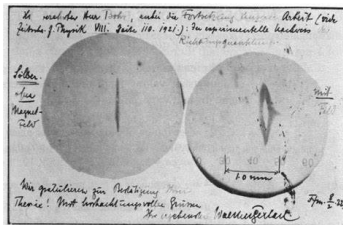
3.1 The Stern-Gerlach Experiment (1922)

- In the Stern-Gerlach experiment, a narrow beam of silver atoms passes through an electromagnet [with a nonuniform $\vec{B}(z) = B(z)\hat{z}$] and then lands on a glass detector plate.



- Silver ($[\text{Kr}]4d^{10}5s^1$) has a magnetic moment, which comes from the spin of the unpaired 5s electron $S = 1/2$. The force on a silver atom is

$$F_z \propto \frac{dB}{dz}.$$

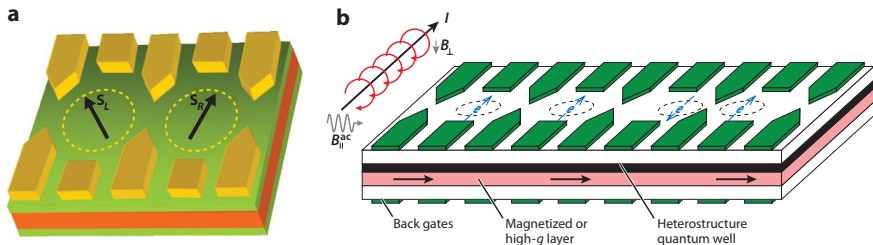


When the electromagnet was turned on, Stern and Gerlach found that the silver atoms formed two distinct branches on the glass plate.

- In the Stern-Gerlach apparatus, if we replace the detector plate and block one of the two beams and pass the other into a second Stern-Gerlach apparatus, will the beam split again? **The answer is no.**
- If spin were a classical quantity that has two values upon measuring, you would argue they form a set, e.g., $\{0, 1\}$ [a set of two values, whose relation (or logic) is Boolean].

- What if we rotate the second apparatus such that the magnetic field (and its gradient) is along the x axis?
- Surprisingly, **the answer is yes**. In particular, we obtain two beams with equal strength in the second case.
- So **spin is not a classical bit**, as the classical (Boolean) description certainly cannot explain the second case with the rotated apparatus.
- In fact, as photon states discussed in Lecture 2, spin states also belong to the 2D complex vector space $\mathbb{C}^2$, so one can encode a qubit in each spin state. We emphasize that the vector space here is not the ordinary 3D real space.

- [Left Panel] Qubits can be encoded in the spin states of quantum dots, each hosting a single electron with spin $1/2$.



- [Right Panel] In the quantum dot array, the qubits in the two right dots are coupled while the others are decoupled. Rotations of individual spins may be achieved by, e.g., electron spin resonance (ESR).

3.2 Spin and Qubit

- The isolated quantum spin ($S = 1/2$) is our second realization of the general class of two-level systems which we call **qubits**, or quantum bits. They are the quantum counterparts of logical bits in classical computers.
- A classical computer operates on strings of zeros and ones, such as 1001001001. Each position in such a string is called a (classical) **bit**, and it contains either a 0 or a 1.
- The abstract bit can be represented by a physical system, such as a capacitor that could be charged (1) or discharged (0), or a magnet whose magnetization could be oriented in two different directions, *up* (0) or *down* (1).

- We can take the two states of a classical bit to be represented by *two orthogonal unit vectors in a two-dimensional space*,

$$|0\rangle \equiv |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle \equiv |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Classically, we either have state $|0\rangle$, or state $|1\rangle$, but not both *simultaneously*.

- In quantum mechanics, however, the two states can be superimposed, i.e., we promote the states of a classical bit to be two basis vectors in $\mathbb{C}^2$.

- As we know, a general state of a qubit is

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad \alpha, \beta \in \mathbb{C}$$

normalized by (the probabilistic interpretation)

$$\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1.$$

- We can parametrize it, alternatively, by

$$|\psi\rangle = \cos(\theta/2) |0\rangle + e^{i\phi} \sin(\theta/2) |1\rangle$$

and represented it by a point on the *Bloch sphere*.

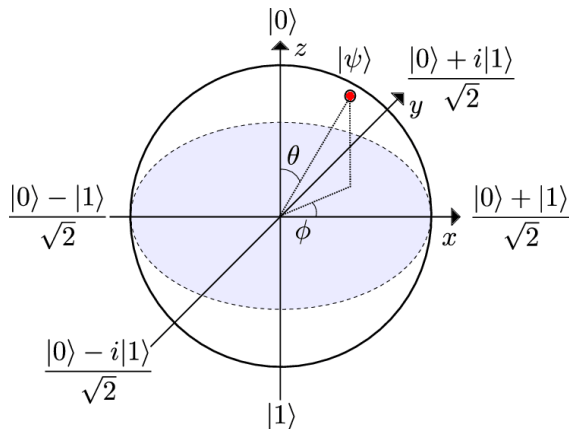


Figure 1: A spin state can be represented by a point on the surface of a sphere of unit radius called *Bloch sphere*.

- Note that we use $\theta/2$, not θ . This guarantees that the relative phase of $|0\rangle$ and $|1\rangle$ is parametrized by ϕ only.
 - If we used θ instead, the states represented by θ, ϕ would be the same as $\pi - \theta, \pi + \phi$, up to an overall phase factor $e^{i\pi} = -1$. So the mapping from spin states to the Bloch sphere would not be 1-to-1 mapping.
- The mapping does not preserve orthogonality. For example, $\langle 0|1\rangle = 0$, but $|0\rangle$ and $|1\rangle$ are represented by unit vectors in opposite directions.

3.3 Linear Operators and Matrices

Take-Home Messages

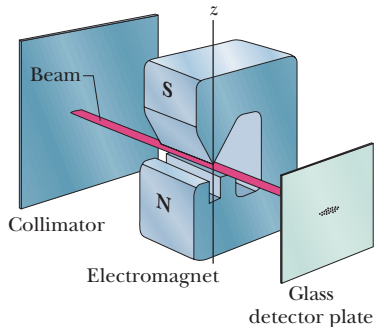
- A quantum state encodes how the system was prepared.
- Physical observables or measurable quantities are described by linear operators, represented by matrices.
- A Hermitian operator can be decomposed with its (real) eigenvalues λ and eigenvectors $|\lambda\rangle$ as

$$M = \sum_{\lambda} \lambda |\lambda\rangle \langle \lambda| = \sum_{\lambda} \lambda P_{\lambda}.$$

Measuring a Quantum Spin

- In classical physics, knowing the state of a system implies knowing everything that is necessary to predict the future of that system.
- Knowing a quantum state, on the other hand, means knowing as much as can be known about how the system was prepared. Perhaps the most difficult things to accept is that quantum mechanics is unavoidably unpredictable. It is a complete calculus of probabilities.
- Our job is to learn and use this calculus (not to challenge it, at least, not before you understand it).

- At the center of the calculus is the idea of an observable or a measurable, i.e., a property one can measure with a proper apparatus.
- To extract information of a spin in a given state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, we make a measurement, which asks whether σ_z (as the outcome of a test) is either $+1$ or -1 .
- A Stern-Gerlach apparatus with magnetic field in the z axis does the job.



Linear Operators

- As we have learned, states in quantum mechanics are mathematically described as vectors in a vector space. But how do we measure the states, mathematically?
- We will introduce an axiom: *physical observables are described by linear operators*.
- A map $M: \mathbb{C}^n \rightarrow \mathbb{C}^n$ is a **linear operator** if

$$M(\mu|x\rangle + \nu|y\rangle) = \mu M|x\rangle + \nu M|y\rangle$$

is satisfied for arbitrary $|x\rangle, |y\rangle \in \mathbb{C}^n$ and $\mu, \nu \in \mathbb{C}$.

- The definition expresses the following facts.
 - M acts on a vector $|a\rangle$ to give $|b\rangle$ in the same space of states:

$$M|a\rangle = |b\rangle.$$

- When M acts on a multiple of an input vector, it gives the same multiple of the output vector:

$$M\zeta|a\rangle = \zeta|b\rangle.$$

- When M acts on a sum of vectors, the results are simply the sum of the output vectors:

$$M\{|a\rangle + |b\rangle\} = M|a\rangle + M|b\rangle.$$

Matrix Representation of Linear Operators

- Given the choice of basis vectors, bra and ket vectors are represented by row and column vectors. If the vector space is N -dimensional, we can choose a set of N orthonormal ket-vectors $|j\rangle$ and their dual bra-vectors $\langle j|$.
- In component form, we can write

$$|a\rangle = \sum_{j=1}^N \alpha_j |j\rangle,$$

$$\langle b| = \sum_{j=1}^N \beta_j \langle j|.$$

- Plugging the substitutions into $M|a\rangle = |b\rangle$, we obtain

$$\sum_{j=1}^N M|j\rangle\alpha_j = \sum_{j=1}^N \beta_j|j\rangle.$$

- The inner product of both sides with a particular basis vector $\langle k|$ gives

$$\sum_{j=1}^N \langle k|M|j\rangle\alpha_j = \sum_{j=1}^N \beta_j\langle k|j\rangle = \beta_k,$$

where the second identity is obtained because $\langle k|j\rangle = \delta_{kj}$.

- Symbolically, we can define an $N \times N$ matrix

$$M = \begin{pmatrix} m_{11} & m_{12} & \cdots & m_{1N} \\ m_{21} & m_{22} & \cdots & m_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ m_{N1} & m_{N2} & \cdots & m_{NN} \end{pmatrix},$$

where $m_{kj} \equiv \langle k|M|j\rangle$ are called matrix elements.

- The inner product of $M|a\rangle = |b\rangle$ with $\langle k|$ becomes

$$\sum_j m_{kj} \alpha_j = \beta_k,$$

for $k = 1, 2, \cdots, N$.

- One recognizes that this is nothing but the matrix multiplication

$$\begin{pmatrix} m_{11} & m_{12} & \cdots & m_{1N} \\ m_{21} & m_{22} & \cdots & m_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ m_{N1} & m_{N2} & \cdots & m_{NN} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_N \end{pmatrix} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_N \end{pmatrix},$$

which can simply be abbreviated as $M|a\rangle = |b\rangle$, in which $M, |a\rangle, |b\rangle$ are now represented by the matrix and two vectors.

Eigenvalues and Eigenvectors

- For the application of a linear operator on a vector, such as $M|a\rangle = |b\rangle$, we do not expect, in general, $|b\rangle = \alpha|a\rangle$, where $\alpha \in \mathbb{C}$.
- However, along certain directions, we find ket-vectors $|\lambda\rangle$ such that

$$M|\lambda\rangle = \lambda|\lambda\rangle,$$

where λ are called **eigenvalues**, while $|\lambda\rangle$ the corresponding **eigenvectors**.

- In general, eigenvalues are complex numbers. For example,

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i \end{pmatrix} = -i \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

- However, observable quantities in quantum mechanics are represented by Hermitian operators, i.e., linear operators that are equal to their own Hermitian conjugate¹.
 - If $M|a\rangle = |b\rangle$, then $\langle a|M^+ = \langle b|$ in general. For Hermitian M , we further have $\langle a|M = \langle b|$.
 - It follows that the eigenvalues of a Hermitian matrix are real.

¹The Hermitian conjugate, denoted by a dagger, is the complex conjugate of a transposed matrix, i.e., $M^+ = [M^T]^*$.

- According to mathematics, the eigenvectors of a Hermitian operator form an orthonormal basis—this serves as a foundation of quantum mechanics.
 - The eigenvectors of a Hermitian operator are a complete set, i.e., any vector the operator can generate can be expanded as a sum of its eigenvectors.
 - If λ_1 and λ_2 are two unequal eigenvalues of a Hermitian operator, then the corresponding eigenvectors are orthogonal.
 - Even if the two eigenvalues are equal (i.e., degenerate), the corresponding eigenvectors can be chosen to be orthogonal.
- Therefore, we often replace an operator by its spectral decomposition (note we insert two identity operators)

$$M = \sum_{\lambda\lambda'} |\lambda\rangle\langle\lambda|M|\lambda'\rangle\langle\lambda'| = \sum_{\lambda} \lambda|\lambda\rangle\langle\lambda| = \sum_{\lambda} \lambda P_{\lambda}.$$

3.4 The Principles of Quantum Mechanics

- 0 States of a system are represented by vectors in a complex vector space.
- 1 The observables or measurable quantities of quantum mechanics are represented by linear operators L .
- 2 The possible results of a measurement are the eigenvalues λ of the operator that represents the observable.
- 3 Unambiguously distinguishable states are represented by orthogonal vectors.
- 4 If $|\psi\rangle$ is a state-vector, and the observable L is measured, the probability to observe λ is $P_\lambda = \langle\psi|\lambda\rangle\langle\lambda|\psi\rangle$, where $|\lambda\rangle$ is the corresponding eigenvector.

Example: Constructing Spin Operators

- Principle 1 states that each component of spin (denoted by σ) is represented by a linear operator.
- To make use of the principles and to yield consistent results to our earlier discussion of spin (and photon polarization), we need to construct linear operators that measure the state along all directions. We begin with σ_z along the z direction.
- Principle 2 states that σ_z has two eigenvalues ± 1 (by definition, as numerical factor can be absorbed) with corresponding eigenvectors, i.e.,

$$\sigma_z|0\rangle = |0\rangle, \quad \sigma_z|1\rangle = -|1\rangle.$$

- Principle 3 emphasizes that $\langle 0|1\rangle = 0$.
- Therefore, as we learned above, the matrix representation of σ_z is

$$\sigma_z = \begin{pmatrix} \langle 0|\sigma_z|0\rangle & \langle 0|\sigma_z|1\rangle \\ \langle 1|\sigma_z|0\rangle & \langle 1|\sigma_z|1\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

where we also use $\langle 0|0\rangle = \langle 1|1\rangle = 1$.

- Principle 4 states that for a spin in $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$,
 - $|\alpha|^2 = \alpha^*\alpha$ is the probability of the spin being up, or that the spin would be measured as $\sigma_z = +1$.
 - Likewise, $|\beta|^2 = \beta^*\beta$ is the probability of the spin being down, or that the spin would be measured as $\sigma_z = -1$.

- How can we construct σ_x and σ_y , then?
- We can also rotate the apparatus to align along x axis and measure the spin component to be either $+1$ or -1 . This means that we can define two basis states by

$$\sigma_x|r\rangle = |r\rangle, \quad \sigma_x|l\rangle = -|l\rangle.$$

- Thus, we need to know, e.g., what α and β are, such that

$$|r\rangle = \alpha|0\rangle + \beta|1\rangle.$$

- In physics, we determine them by experimental measurements, not mathematical axioms.

- First, we need to prepare spins in the state $|r\rangle$. This can be done by passing particles in arbitrary spin states through a Stern-Gerlach apparatus oriented along x direction.
 - In general, the beam will be split into two. The one on the right contains particles with spin in the state $|r\rangle$.
 - We block the beam on the left.
- Next, we pass the beam of spin $|r\rangle$ particles through a Stern-Gerlach apparatus oriented along z direction.
 - Experiments find that the beam is split into two with equal probabilities. Thus, we must have $|\alpha|^2 = |\beta|^2 = 1/2$.
- Therefore, experiments tell us that

$$|r\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{e^{i\phi}}{\sqrt{2}}|1\rangle.$$

- There is some ambiguity in choosing the phase ϕ , but it is nothing more than the ambiguity in our choice of the exact direction for the x axis. We can simply choose $\phi = 0$, so

$$|r\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle.$$

- Using $\langle r|l\rangle = 0$ and $\langle l|l\rangle = 1$, we find

$$|l\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle.$$

- There can be an overall phase factor in $|r\rangle$ and $|l\rangle$, but it is not measurable.

- The matrix representation of σ_x must, then, satisfy

$$\begin{pmatrix} (\sigma_x)_{00} & (\sigma_x)_{01} \\ (\sigma_x)_{10} & (\sigma_x)_{11} \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix},$$

$$\begin{pmatrix} (\sigma_x)_{00} & (\sigma_x)_{01} \\ (\sigma_x)_{10} & (\sigma_x)_{11} \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} = - \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix},$$

- The solution to the two matrix equations is

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

- In a similar, but more complicated way, we can obtain

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},$$

whose eigenvectors, with eigenvalues ± 1 , are

$$|i\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle,$$

$$|o\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle.$$

Along an Arbitrary Direction

- The three spin operators, represented by

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

are known as **Pauli matrices**.

- It is tempting to write

$$\vec{\sigma} = \sigma_x \hat{x} + \sigma_y \hat{y} + \sigma_z \hat{z}.$$

This is correct. In fact, spin is a 3-vector operator, as its measurement results depend on spatial direction.

- It is good time to distinguish the two notions of vectors.
 - We live in a 3D world. Spin, like displacement and momentum, is a 3-vector. It has operator (not scalar) components along the three spatial directions.
 - The quantum state of a spin is a state-vector in a 2D complex vector spaces (of spin states).
- Therefore, the relevant operator for measuring the spin along an arbitrary direction $\hat{n} = (n_x, n_y, n_z)$ is

$$\sigma_n = \vec{\sigma} \cdot \hat{n} = \sigma_x n_x + \sigma_y n_y + \sigma_z n_z = \begin{pmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{pmatrix}.$$

- One can check that the eigenvalues, or the measurement results, are, again, ± 1 .

Irreversibility of Measurement

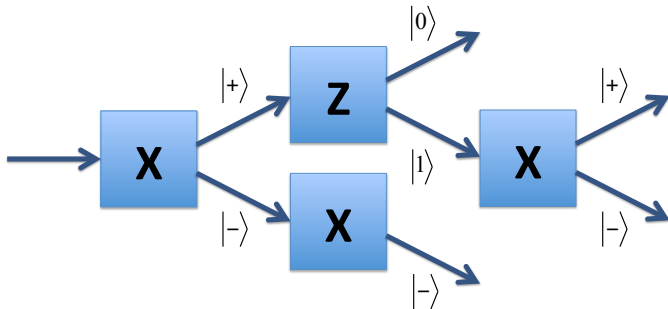
- Quantum measurement is *irreversible*. We say that the pre-measurement state *collapses or reduces* to the post-measurement state, as a consequence of the measurement (the Copenhagen interpretation).

$$|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle \rightarrow \boxed{\overset{x}{M}} \quad |x\rangle \quad p = |\alpha_x|^2$$

- Note when an operator acts on a state-vector, it produces a new state-vector in general. The new vector has nothing to do with the post-measurement state.²

²Though, we can use it to calculate the average or expectation value of the measurable quantity (Lecture 4).

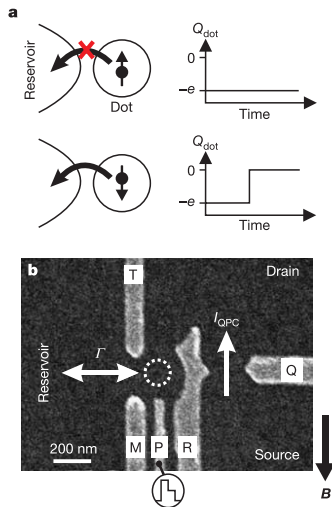
- After an ideal measurement, the system is in one of the eigenstates. Measurement is like blocking all but one of the paths. **What if you do not block it?**
 - Along other paths, you either get the particle, or you do not.
 - You do not know when you get it, but you know the probability of getting it.



Measuring Electron Spin in a Quantum Dot³

Single-shot read-out

- Gate voltage pulse shifts the Zeeman-split spin levels (spin-up and spin-down) relative to the Fermi energy in the reservoir.
- Spin-down electron can tunnel out, while a spin-up electron remains bound, enabling the single-shot spin-to-charge conversion measurement.



³J. M. Elzerman et al., Nature 430, 431 (2004).

Discussion Questions:

- 1 What is meant by “single-shot measurement”?
- 2 Why do electrons with different spins exhibit different tunneling behaviors?
- 3 Does “single-shot measurement” imply that the measurement time is very short? What determines the measurement time?
- 4 What is the primary reason for the higher misidentification rate (lower fidelity) for the “spin-down” state?
- 5 Has the quantum state of the electron collapsed? Is the electron still confined after the measurement?

Summary and Outlook

- A quantum system evolves by unitary development (Lecture 5) between measurements and by state (or wave function) collapse when measurements take place.
- From quantum information point of view, this is schematically how a quantum computer operates.
- Given an initial state $|\psi_0\rangle$, the system evolves as

$$|\psi(t)\rangle = U(t)|\psi_0\rangle,$$

where $U(t)$ is the time-evolution operator (Lecture 5).

- Suppose a measurement is associated with a Hermitian operator L with eigenvalues λ and eigenvectors $|\lambda\rangle$. We can predict the probabilities for each possible outcome of the measurement as a function of time

$$P_\lambda(t) = |\langle\lambda|\psi(t)\rangle|^2.$$

- In a real experiment at time t , the apparatus measures a value λ with probability $P_\lambda(t)$ and leaves the system in the corresponding eigenstate $|\lambda\rangle$.

References

- L. Susskind and A. Friedman, Quantum Mechanics (Basic Books, 2014), Chaps. 3-4.
- M. Nakahara and T. Ohmi, Quantum Computing: From Linear Algebra to Physical Realizations (CRC Press, 2008), Chap. 1.

Homework

P3-1. Verify that the Pauli matrices $\{\sigma_x, \sigma_y, \sigma_z\}$ satisfy the following algebraic relations:

- a. $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = 1.$
- b. $\sigma_x \sigma_y = -\sigma_y \sigma_x = i\sigma_z, \sigma_y \sigma_z = -\sigma_z \sigma_y = i\sigma_x,$ and $\sigma_z \sigma_x = -\sigma_x \sigma_z = i\sigma_y.$ Therefore,

$$\frac{\vec{\sigma}}{2} \times \frac{\vec{\sigma}}{2} = i\hbar \frac{\vec{\sigma}}{2}.$$

- c. $(\vec{a} \cdot \vec{\sigma})(\vec{b} \cdot \vec{\sigma}) = (\vec{a} \cdot \vec{b})I + i(\vec{a} \times \vec{b}) \cdot \vec{\sigma},$ where I is the 2×2 identity matrix.

P3-2. The general spin-1/2 (or qubit) state can be expressed by

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle.$$

- (a) Show that this state is an eigenvector of the operator $\hat{n} \cdot \vec{\sigma}$, where $\hat{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$. What is the corresponding eigenvalue?
- (b) Calculate the expectation value of the Pauli matrices $\langle \sigma_x \rangle$, $\langle \sigma_y \rangle$, $\langle \sigma_z \rangle$ in this state, i.e., $\langle \sigma_i \rangle \equiv \langle \psi | \sigma_i | \psi \rangle$, $i = x, y, z$. Show that $\langle \vec{\sigma} \rangle$ points along $\hat{n}$.

P3-3. The spin-1/2 representation ($\vec{S} = \vec{\sigma}/2$) is the simplest nontrivial, and defining, representation of the $SU(2)$ algebra

$$\vec{S} \times \vec{S} = i\hbar\vec{S},$$

which is equivalent to the angular momentum algebra in quantum mechanics. In particular, the spin-1 representation reads

$$S_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad S_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad S_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Calculate the eigenvalues and eigenvectors of S_x and S_y .